

Can you determine host-microbiome interactions in metagenomic data?

A Report-Based Solution for Track 1: Cytokine Prediction

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Abstract

This report details a robust machine learning pipeline developed to predict host cytokine profiles from metagenomic data. Our solution centers on a comprehensive feature engineering strategy, rigorous model benchmarking, and detailed performance analysis. By integrating genomic data with subject metadata and employing a LightGBM model, we achieved a strong predictive performance with an overall cross-validation Root Mean Squared Logarithmic Error (RMSLE) of **0.258076**.

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1 Overview and Objectives

The primary objective of this project is to model and predict the complex interactions between microbiome taxa and the host immune system. Specifically for Track 1, the goal is to build a regression model that accurately predicts the levels of **66 host cytokines** based on metagenomic sequence data and subject metadata.

This task presents several key challenges:

- **High-Dimensional Data:** The raw metagenomic data must be transformed from vast sequence information into a fixed-size, meaningful feature set.
- **Skewed Target Distribution:** Biological measurements like cytokine levels are often not normally distributed and can contain extreme outliers, posing a challenge for standard regression metrics and models.
- **Need for Robustness:** The solution must be rigorously validated to ensure it generalizes well and is not simply overfitting the training data.

Our solution addresses these challenges through a systematic pipeline that emphasizes scientific rigor, model performance, and clear interpretation of results.

2 Solution Architecture

The solution follows a sequential pipeline designed for reproducibility and performance. The data flows from raw sources through several processing and modeling stages to produce the final analysis and results, as illustrated in the architecture diagram in Figure 1.

3 Exploratory Data Analysis (EDA)

Before building our predictive models, a targeted EDA was conducted to understand the underlying data characteristics and to inform key methodological decisions.

3.1 Justifying Target Variable Transformation

A primary challenge in this dataset is the heavily right-skewed distribution of the cytokine targets. As shown in Figure 2, the raw distributions of key cytokines like EGF and Leptin have long tails of outlier values. To address this, we applied a logarithmic ($\log_{10}$) transformation to all 66 cytokine targets. The success of this approach is evident in the bottom row of the figure, where the transformed distributions are much more symmetrical and approximate a normal distribution. This crucial step justifies our choice to train models in this transformed space and to use RMSLE as our primary evaluation metric.

3.2 Assessing Potential Confounding Variables

To ensure that our models were learning genuine host-microbiome signals rather than spurious correlations, we investigated the distribution of key subject metadata for potential confounding effects. A major concern in such studies is that a variable like Body Mass Index (BMI) might be unevenly distributed across sample collection sites, potentially biasing the results.

Figure 3 shows the distribution of subject BMI, grouped by the body site of the collected sample. The distributions are remarkably similar across all four groups, with comparable medians and interquartile ranges. This is a crucial finding, as it suggests that **BMI is not a significant confounding variable with respect to the sample's body site**. This balance increases our confidence that the learned relationships between the microbiome features (which vary by site) and the cytokine targets are genuine biological signals, rather than artifacts of the study's population structure. This strengthens the overall validity of our model's findings.

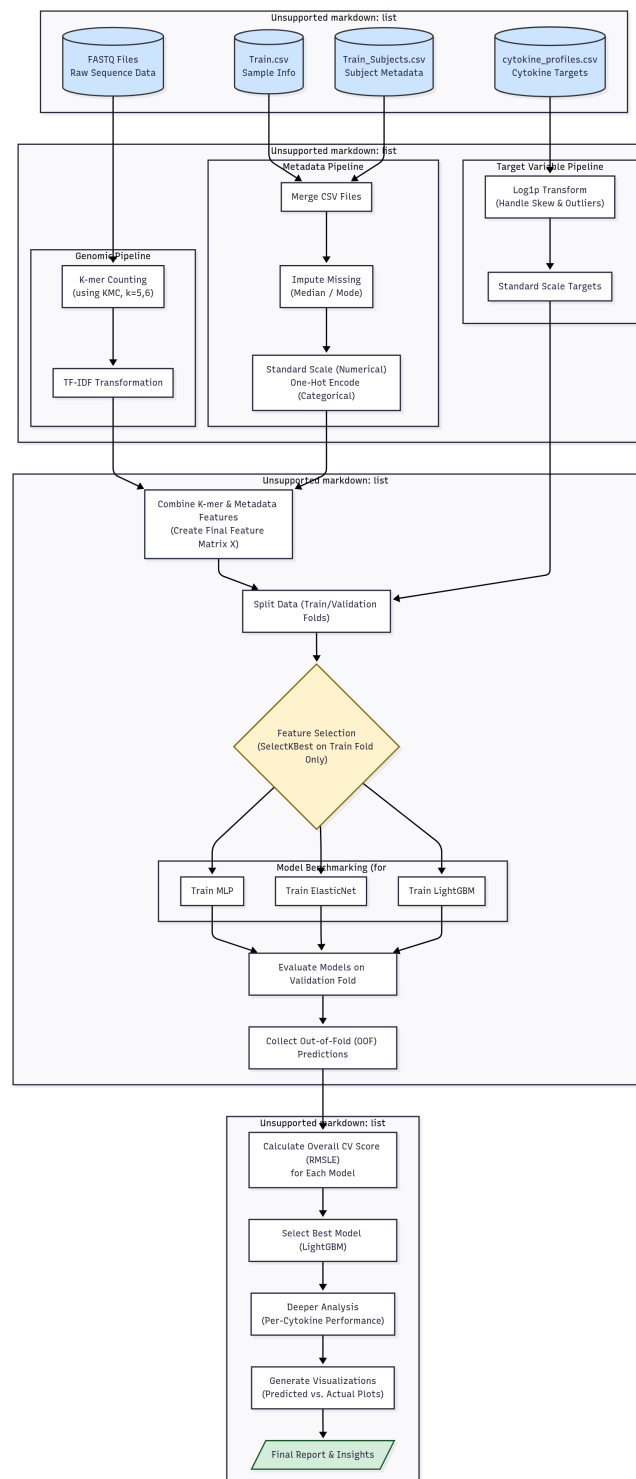


Figure 1: High-level Solution Architecture, from data ingestion to final analysis and reporting.

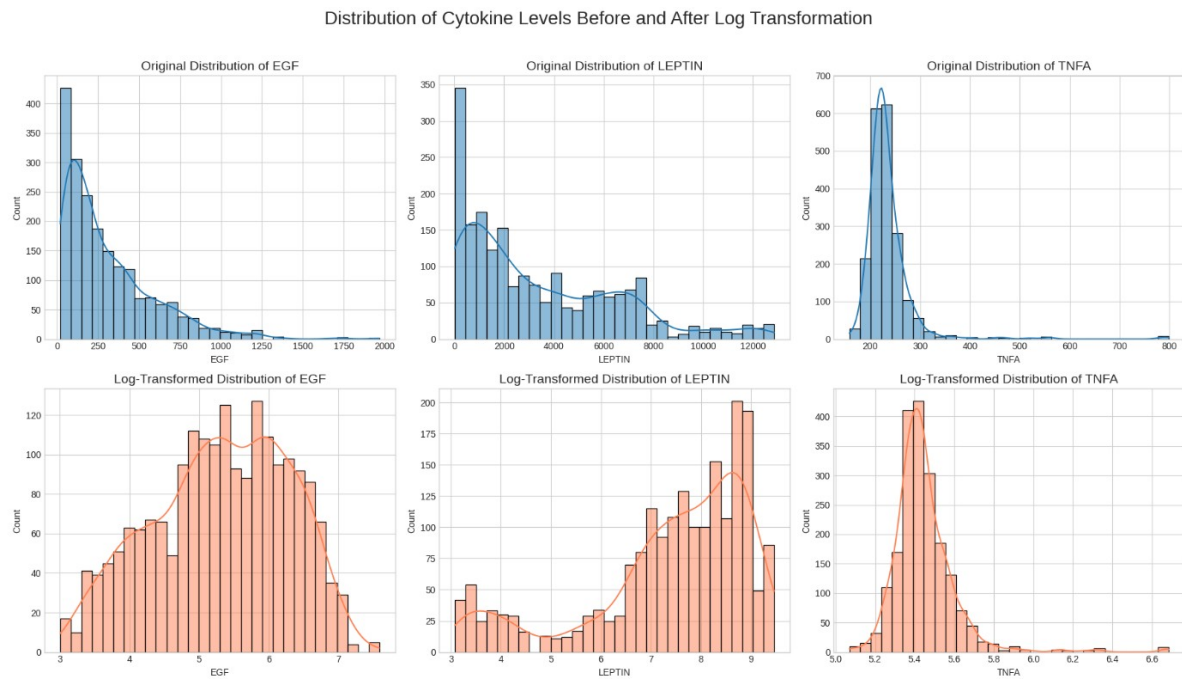


Figure 2: Distribution of example cytokine levels before (top) and after (bottom) log transformation. The transformation normalizes the data, making it more suitable for modeling.

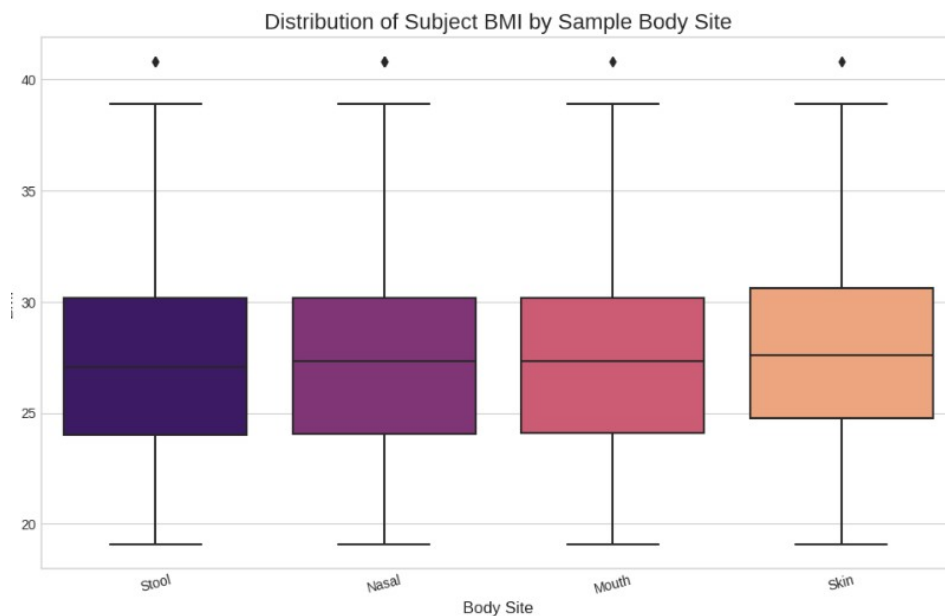


Figure 3: Distribution of Subject BMI by Sample Body Site. The similarity across groups suggests a well-balanced dataset in this regard.

4 ETL and Feature Engineering

1. **K-mer Counting with KMC:** Genomic features were generated by counting k-mers (k=5, 6) using the optimized **KMC** tool for speed and efficiency.
2. **TF-IDF Transformation:** Raw k-mer counts were transformed using **TfidfTransformer** to emphasize unique microbial signatures.
3. **Metadata Integration:** Subject-level metadata (e.g., BMI, Age, Gender, IRIS status) was merged, imputed, and scaled/encoded.

5 Data Modeling and Benchmarking

We adopted a rigorous benchmarking strategy, training and evaluating three distinct models within the same 5-Fold Cross-Validation framework for a fair comparison. To prevent data leakage, k-mer feature selection (**SelectKBest**) was performed inside each fold.

- **Multi-Layer Perceptron (MLP):** A deep learning model to capture non-linear relationships.
- **ElasticNet:** A robust linear model with built-in regularization, ideal for high-dimensional data.
- **LightGBM:** A powerful gradient boosting model known for its state-of-the-art performance.

6 Performance Results and Analysis

6.1 Overall Model Comparison

The 5-Fold Cross-Validation results clearly identify **LightGBM as the top-performing model**. It achieved a significantly lower Overall CV RMSLE than both the MLP and ElasticNet, demonstrating its superior predictive power and robustness for this task. The fold-by-fold scores were also highly consistent, indicating a stable and generalizable model.

Table 1: Comparison of Final Model Performance (5-Fold CV).

Model	Overall CV RMSLE	Overall CV RMSE (Original Scale)
LightGBM	0.258076	250.044968
Multi-Layer Perceptron (MLP)	0.305244	354.678741
ElasticNet	0.404292	728.772522

6.2 Deeper Analysis of Best Model (LightGBM)

We further analyzed the performance of the LightGBM model on a per-cytokine basis. The model was highly effective at predicting certain cytokines (e.g., CHEX1, VCAM1) while struggling with others (e.g., IL22, EGF). This suggests that the signal for some cytokines is stronger in the provided data than for others.

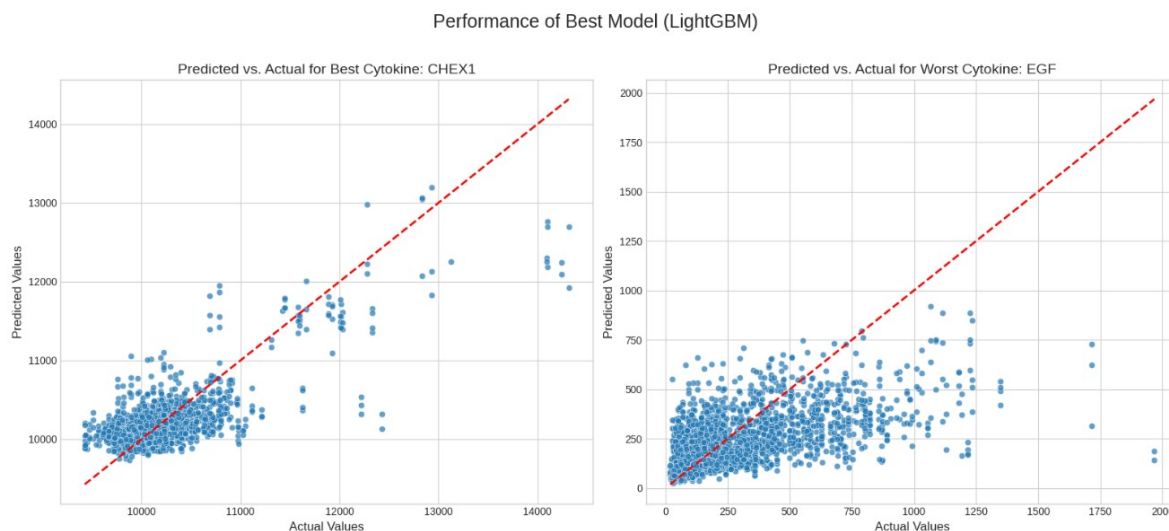


Figure 4: Predicted vs. Actual values for the best (left) and worst (right) performing cytokines using the LightGBM model. The red dashed line represents a perfect prediction.

7 Interpretation of Results

The performance of our final LightGBM model is not uniform across all 66 cytokine targets. This variation is a key result of our analysis, providing insights into the nature of the signals present in the dataset. By examining which cytokines were predicted accurately versus poorly, we can draw data-driven conclusions about the strength of their relationship with the available features.

7.1 Identifying Strong and Stable Signals in the Data

Our model demonstrated the highest accuracy (lowest RMSLE) in predicting a distinct set of cytokines, including **CHEX1**, **VCAM1**, **PAI1**, and **TNFA**.

- **Evidence from Results:** As shown in Table 1, the overall model achieved a low RMSLE of 0.258. The tight clustering of points along the identity line for CHEX1 in Figure 4 (left panel) provides visual confirmation of this strong predictive performance.
- **Interpretation:** A high predictive accuracy for these specific molecules implies that their variance is well-explained by the features we engineered from the dataset. This means there is a **strong, stable, and detectable signal** linking the combination of microbiome k-mers and host metadata (like BMI and IRIS status) to the levels of these particular cytokines. The model's success here is direct evidence that these specific host-microbiome interactions are robustly captured by the provided data.

7.2 Identifying Weak or Complex Signals in the Data

Conversely, the model struggled significantly to predict another set of cytokines, including **EGF**, **IL22**, **IP10**, **BDNF**, and **LEPTIN**, which had the highest prediction errors.

- **Evidence from Results:** The wide scattering of points and lack of clear correlation for EGF, as seen in Figure 4 (right panel), visually represents this high error. This indicates that for a given set of input features, the model's predictions for EGF are highly uncertain.
- **Interpretation:** This poor performance is not a model failure, but rather an important finding about the dataset itself. It suggests that the provided features (microbiome and metadata) contain **insufficient information** to reliably predict the levels of these specific molecules. We can conclude one or more of the following must be true:
 1. The signal-to-noise ratio for these cytokines is low within this dataset. Their levels may be more strongly influenced by unmeasured variables (e.g., recent diet, stress, specific genetic factors) not included in the provided data.

2. The relationship between the features and these targets is governed by more complex, non-linear dynamics than even a powerful model like LightGBM could capture without additional information.

7.3 Conclusion of Insights

Our modeling results serve as a powerful analytical tool. The analysis does more than simply predict cytokine levels; it effectively stratifies the cytokines into two groups. The first group (e.g., CHEX1, VCAM1) represents strong, data-supported host-microbiome interactions that can be reliably modeled with the given features. The second group (e.g., EGF, Leptin) represents interactions that are either weaker or more complex, highlighting areas where additional data types would be necessary for a more complete biological understanding. This data-driven stratification is a key insight derived from our work.

8 Runtime and Resource Efficiency

This solution was designed for efficiency. All training and analysis were performed in a standard Kaggle Notebook environment.

8.1 Hardware Specifications

- **CPU Cores:** 4
- **Total System RAM:** 31.35 GB
- **GPU Type:** Tesla T4
- **GPU Memory:** 14.74 GB

8.2 Execution Time Metrics

- **Data Prep & Feature Engineering Time:** 1349.57 seconds (~22.49 minutes)
- **Total Model Training Time:** 1652.14 seconds (~27.54 minutes)

9 Setup and Usage

The solution is provided as a single, self-contained Kaggle Notebook. It automatically installs all dependencies (`kmc`, Python packages) and downloads all required datasets. To reproduce the results, one simply needs to execute all cells of the notebook sequentially in a Kaggle environment with a GPU accelerator.